Manuscript outline

Revised Figures 2–4: challenge signatures, separation and noisy polarity

Completed revisions, detailed legends and linked comprehensive evidence

20 October 2026 · Completed Figures 2–4 revisions with stated scientific qualifications

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes how established polarity is organized and replaced; these changes may provide a physical basis for greater migration persistence. The theoretical contribution tests size under unchanged local kinetic parameters. Length differs between model cells to represent WGD-associated enlargement, rather than elongation during a turn. Existing migration observations supply empirical motivation; new paired WGD motility results retain a substantial, explicitly reserved contribution.

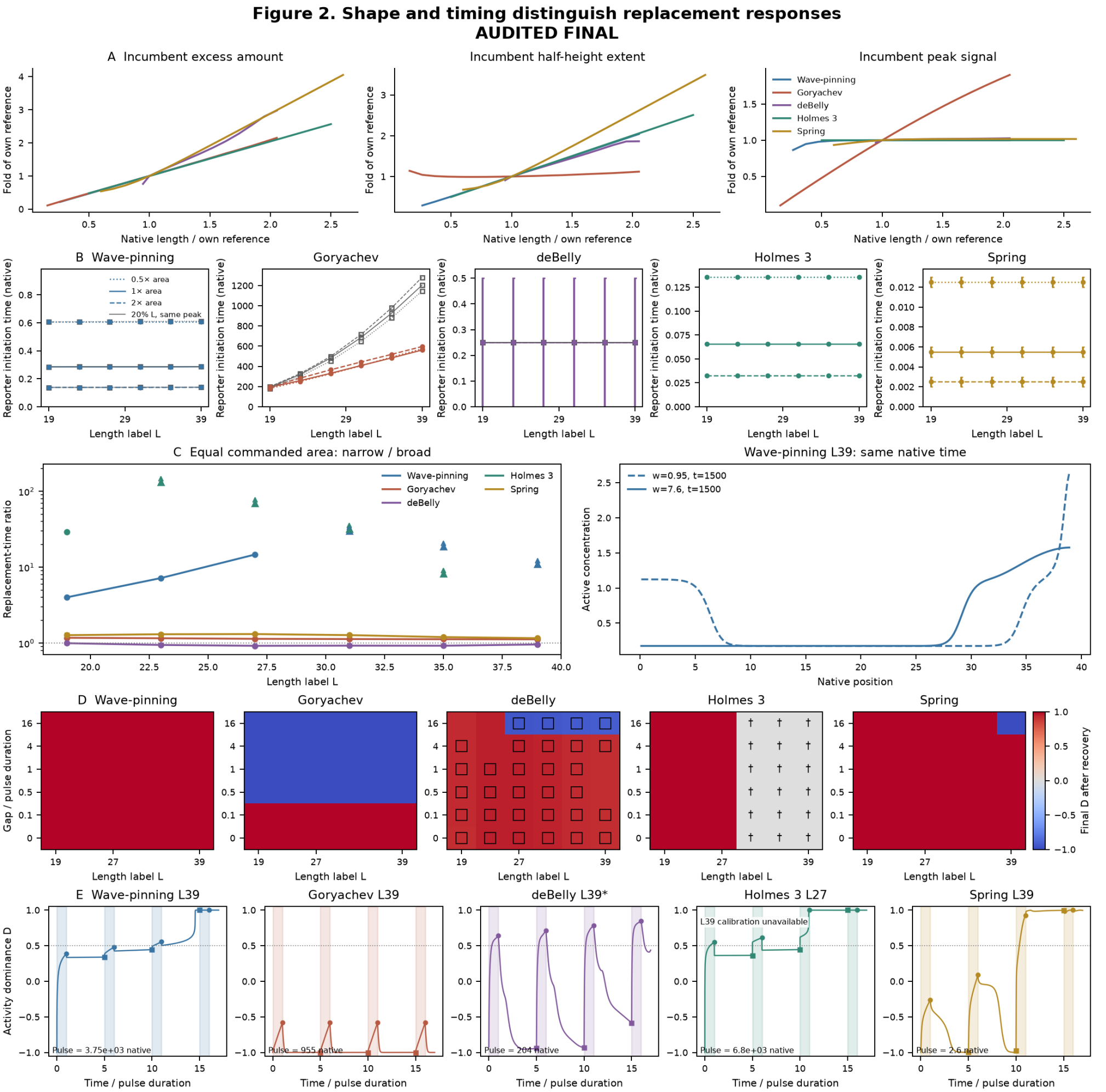
Figures 2–4 test how challenge shape and timing separate response mechanisms, how physical separation affects competition at fixed size, and how replicated fluctuating input changes polarity-direction persistence and spatial organization. Figure 2 reports equal-area widths and pulse trains. Figure 3 distinguishes cue-driven handover, cue-free resolution and directly measured activity direction. Figure 4 follows its completed hypothesis, coverage and sensitivity decisions. Detailed native histories and every finite-horizon or numerical qualification remain in the linked supplements.

This is a working manuscript outline. Figures 1 and 5 and the reserved Figure 6 experimental section retain their preceding content. Activity polarity, modelled replacement, observed migration and conditional OU velocity memory remain distinct readouts; their biological connection is an experimental hypothesis.

Proposed main figures

- Figure 2. Shape and timing distinguish replacement responses
- Figure 3. Separation affects native competition differently across models
- A size advantage in heading persistence can coexist with multiple active regions.

Figure 2. Shape and timing distinguish replacement responses



Model	Trace L	Initiation L39/L19	Narrow / broad, L19 → L39	Pulse 1 → gap 1: D	Final D
Wave-pinning	L39	1–1.01	3.99–4.05 → >11.1	0.39 → 0.34	1.00
Goryachev	L39	2.87–2.92	1.17–1.18 → 1.12–1.13	-0.58 → -1.00	-1.00
deBelly*	L39	both in (0, 0.5]	0.962–1.04 → 0.955–0.97	0.64 → -0.94	0.92
Holmes 3	L27	0.985–1.02	29–29.1 → uninformative	0.55 → 0.36	1.00
Spring	L39	0.833–1.2	1.2–1.36 → 1.15–1.18	-0.26 → -1.00	1.00

Shape: beam width at fixed input area • Timing: equal total ON pulse trains • Reporter: activity share after each gap

↑ triangles: finite-cap ratio lower bounds; bounds cannot be ranked. †: no calibration; ×: low activity; □: failed field tail.

* deBelly later-gap timing and endpoint fields have numerical qualifications (S5). Native clocks differ; optical dose is uncalibrated.

Five-model shape and timing signature assays

Figure 2. Shape and timing distinguish replacement responses

Test whether shape and timing of a measured challenge distinguish explanations for size-dependent replacement resistance.

Detailed legend

A, Retained organization of all 97 qualified basal states. Excess amount is integrated inside the interpolated basal half-height core. Amount, extent and native peak signal are divided by each model's own reference values; Spring retains its physical domain and native resting-length convention.

B, Reporter initiation: first cue-site positive-excess peak reaching 10% of the exact L19 basal excess peak, held fixed within each model across all lengths, widths and strengths. The fixed physical-width raised-cosine cue has support 3.8 chemical units or 0.1 Spring units. Dotted, solid and dashed coloured curves have 0.5, 1 and 2 times the reference commanded spatial input area. Grey open-square curves retain the same peak at relative support 20% of length for the three focal models; this diagnostic changes spatial area with length. Bars are saved-sample crossing brackets. deBelly events in the first native update remain in the interval (0, 0.5]; substep interpolation is descriptive only. Open diamonds at zero denote a footprint already above the reporter threshold before cue application. Native clocks are shown separately.

C, Time to eligible D ≥ 0.5 during continuous cue ON for width 0.95 divided by width 7.6 (Spring widths 0.025 and 0.2), at the central fixed commanded spatial area equal to native reference peak times 3.8 (Spring times 0.1). Bars propagate both sampled crossing brackets; upward triangles and arrows are strict finite-cap lower bounds when the narrow arm remains below replacement at its native horizon. Censored bounds are not finite replacement times and cannot be ranked as measured ratios. The profile inset compares exact native L39 fields at the predeclared saved time nearest 1500 with one common concentration scale.

D, Final activity dominance after four equal pulses with total ON 1.5 times each model–length's postwithdrawal commitment calibration, then 10,000 native cue-free units (4000 for deBelly). Gap durations are multiples of one pulse. E, Four pulses with gaps four times one pulse at L39 where calibrated; otherwise the largest calibrated length is used. This deterministic fallback shows Holmes 3 at L27, explicitly labelled, because L39 has no sampled continuous-cue commitment through 100,000 ON units. Selection depends on calibration availability, not train fate. Shading marks cue ON; circles and squares mark pulse and gap endpoints. Time is divided by each model's actual pulse duration, printed in native units. F, The trace-length column preserves this exception beside the numerical signature summary. Shared responses at calibrated lengths do not establish a matched-length comparison at L39.

Dominance D is the positive active-signal excess in the challenger half minus that in the incumbent half, divided by their sum. The instantaneous profile minimum is the common floor. Eligible D = 0.5 means 75% challenger share; low-activity states require explicit amount and contrast gates and receive no invented D. A square marks a failed finite

Figure 2. Shape and timing distinguish replacement responses · continued

native-field tail independently of final fate. Grey dagger cells have no sampled continuous-cue commitment through the explicitly searched finite cap, so no calibration duration or train trajectory is assigned. Plain grey cells lack a completed independent audit; neither is censoring of an executed train.

Commitment calibration is the shortest sampled continuous ON duration, bracketed by native-duration bisection, that yields eligible D ≥ 0.5 after withdrawal and the declared model-native recovery period. The finite native-field tail is evaluated independently and does not redefine scalar replacement. Native clocks differ between models; no absolute cross-model time ranking is used.

The deBelly trace and summary carry an asterisk because later-gap commitment timing is numerically sensitive. At L39, the third-gap endpoint is D = −0.586 with the native 0.5-unit update and D = +0.190 in the separately qualified 0.25-unit companion. The first two gaps still collapse and final replacement agrees, but later-gap timing and field-tail status are not resolution invariant. The L39 401-cell train also has native endpoint-field sensitivity despite phase-D and fate agreement. S5 retains exact-native-start and separately qualified quarter-step preparations, every endpoint, native-field errors and all gate outcomes. Noisy deBelly ensembles use the native 0.5-unit update and remain conditional on that numerical scheme.

All five models keep their native kinetics. Commanded spatial input area per ON time is not calibrated photon dose or realized activated molecular amount. A width penalty or gap retention is an observable signature, not a causal allocation to a particular feedback term or unique molecular identification. Holmes 3 and Spring determine whether supported signatures are shared. Polarity replacement is distinct from observed migration and conditional OU persistence estimates.

S3 reports conditional retention through weakly driven gaps, separately from final replacement. At the third gap with gap duration 16 times one pulse, using the predeclared 80% raw-share criterion with D ≥ 0, counts at mean input 0.5% and 2% of native peak are Wave-pinning L19: 11/12 and 8/12; Wave-pinning L29: 12/12 and 12/12; Wave-pinning L39: 12/12 and 12/12; deBelly L19: 12/12 and 11/12; deBelly L29: 0/12 and 0/12; deBelly L39: 0/12 and 0/12. These are independent-seed denominators, with exact 95% intervals shown in S3; the three gaps of one trajectory are not independent replicates. Mean and zero controls remain visible. Weak describes instantaneous bounded input (maximum 1% or 4% of native peak); at L39 with gap multiple 16 and mean input 2%, whole-domain gap exposure is about 2.46 times targeted ON exposure. This commanded native-coordinate coefficient is not photon dose or realized activation. deBelly retention remains conditional on the native 0.5-unit update.

Narrative points and conclusions

- Narrow-to-broad replacement-time ratios at L19 and L39 are Wave-pinning: 3.99–4.05 → >11.1; Goryachev: 1.17–1.18 → 1.12–1.13; deBelly*: 0.962–1.04 → 0.955–0.97;

Figure 2. Shape and timing distinguish replacement responses · continued

Holmes 3: 29–29.1 → uninformative; Spring: 1.2–1.36 → 1.15–1.18. These are sampled bounds, not fitted scaling laws.

- Activity-share trajectories and their exact gap endpoints test whether an early challenged state persists through cue-free gaps. A later retained replacement does not establish that early activity itself persisted.

Proposed measurements and experimental follow-up

Vary physical cue support at fixed commanded spatial input area and measure time to eligible75% challenger activity share.

Apply equal-total-ON pulse trains, measure activity share at every pulse and gap endpoint, then report postwithdrawal fate and field stability independently.

Scope, limitations and pending validation

Native-to-optical calibration is absent. Native clocks, reporter species, amount scales and stimulus channels differ across models.

Observable signatures do not uniquely identify a molecular mechanism or allocate causation to one feedback term. All five models determine whether responses are shared.

deBelly early activity collapse and final replacement survive the qualified quarter-step check, but the third-gap dominance sign and finite-field-tail status change; the L39 refined-mesh train also has endpoint-field sensitivity. Later commitment timing and native-step noise probabilities are not claimed to be numerically invariant.

Finite native-field tails qualify only the declared horizon; deterministic activity replacement does not measure migration persistence.

Sources and reproducibility

Independent coverage and scientific source receipts

Sampled width-ratio bounds

Selected pulse/gap endpoint measurements

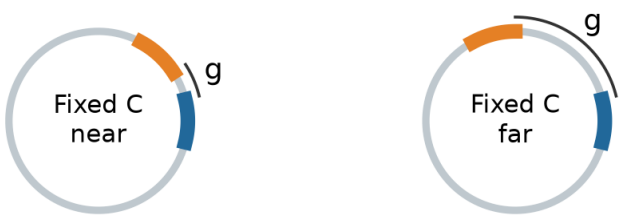
Material deBelly temporal qualification and all phase gates

Complete bounded-noise retention and finite-seed uncertainty

All numerical companions and material qualifications

Figure 3. Separation and native competition · image 1 of 2

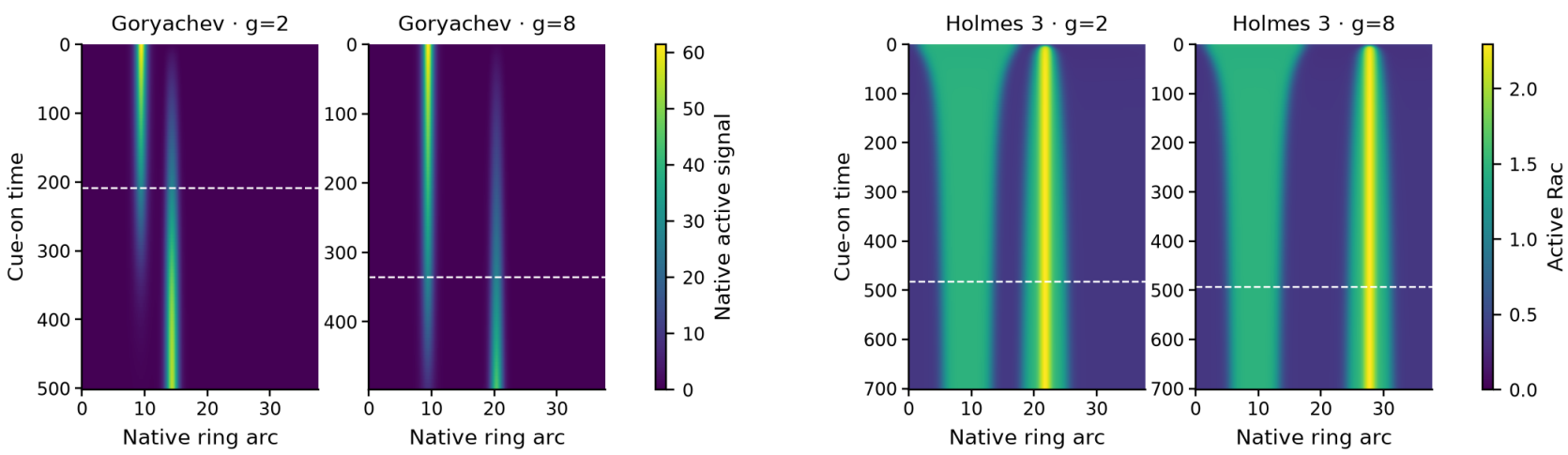
Figure 3 · Separation affects native competition differently across models



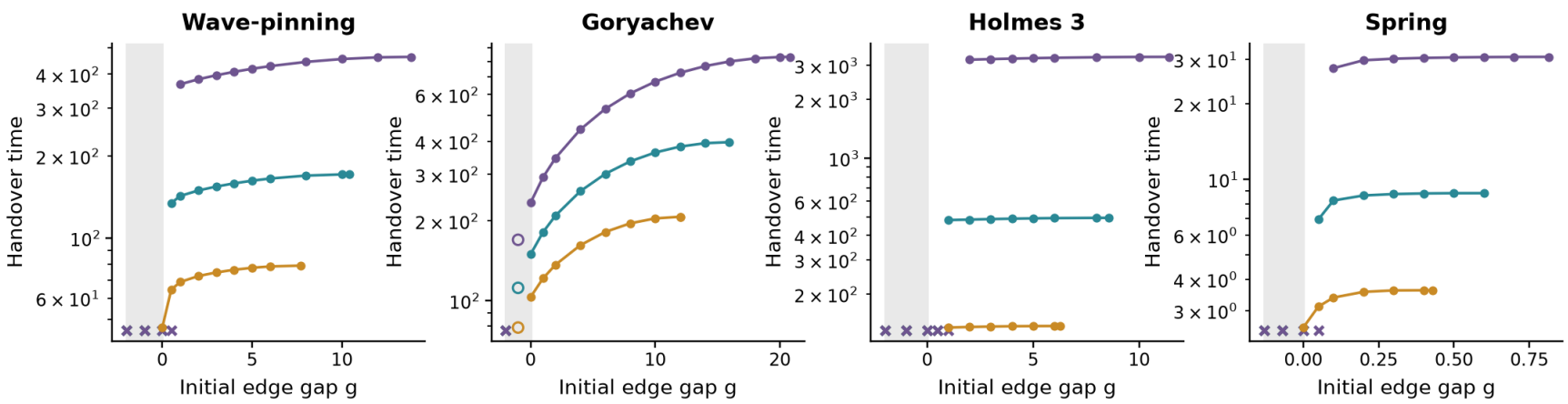
A Rotate one native cue at fixed size

g: initial incumbent 10% edge → cue near edge
A1: continuous cue → complete-region handover
A2: own-handover-scaled pulse → cue-free fate
B: seed ratio 0.5 / 1 / 2 → cue-free peak resolution
12 fresh qualified basals; native kinetics unchanged

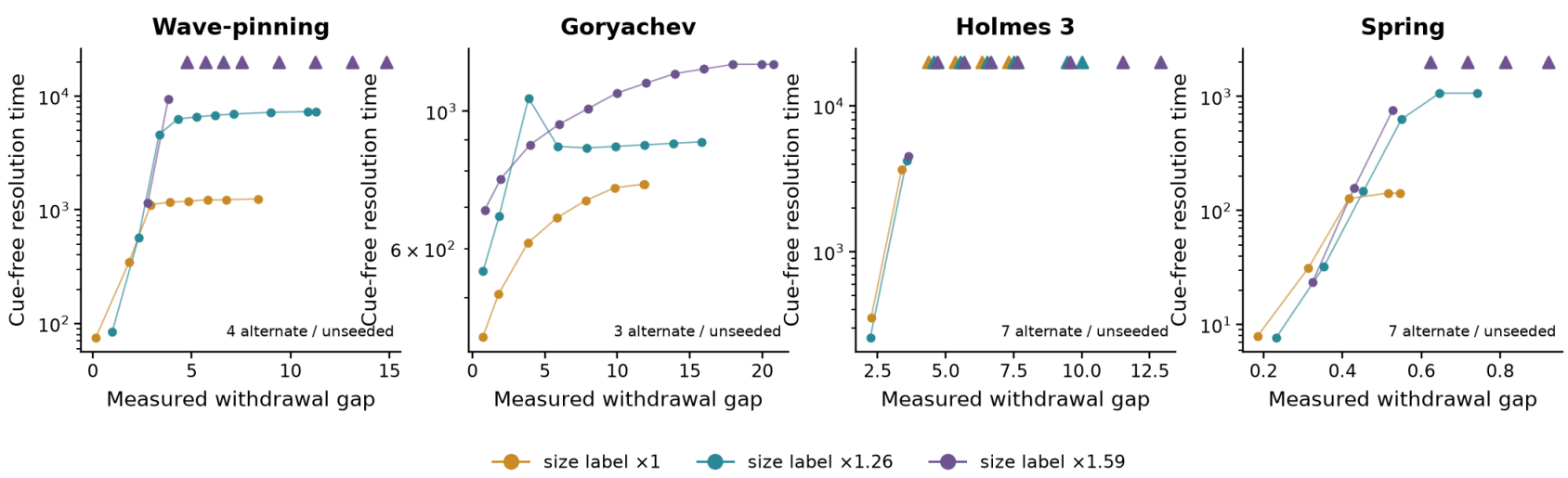
B Same-size native histories · white line: complete-region handover



C1 Driven handover · open symbols: borderline placements



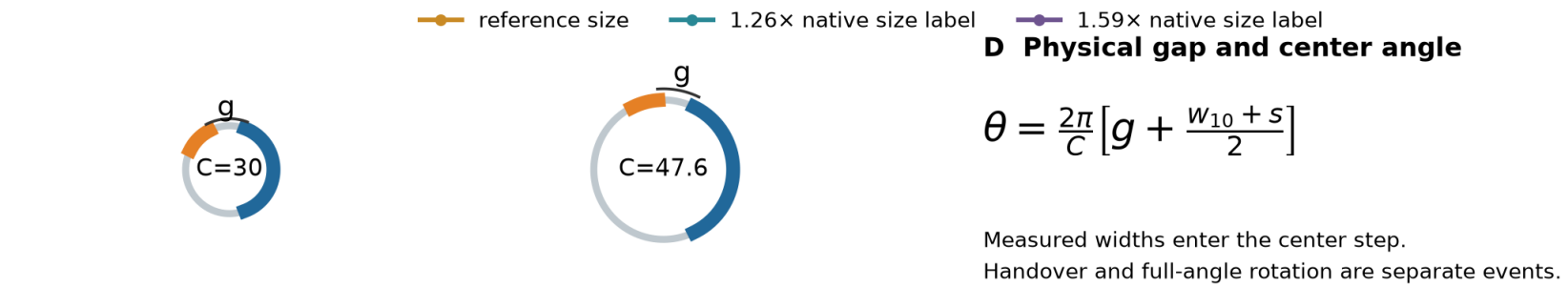
C2 Cue-free resolution · ratio 1 · triangles retain finite-cap lower bounds

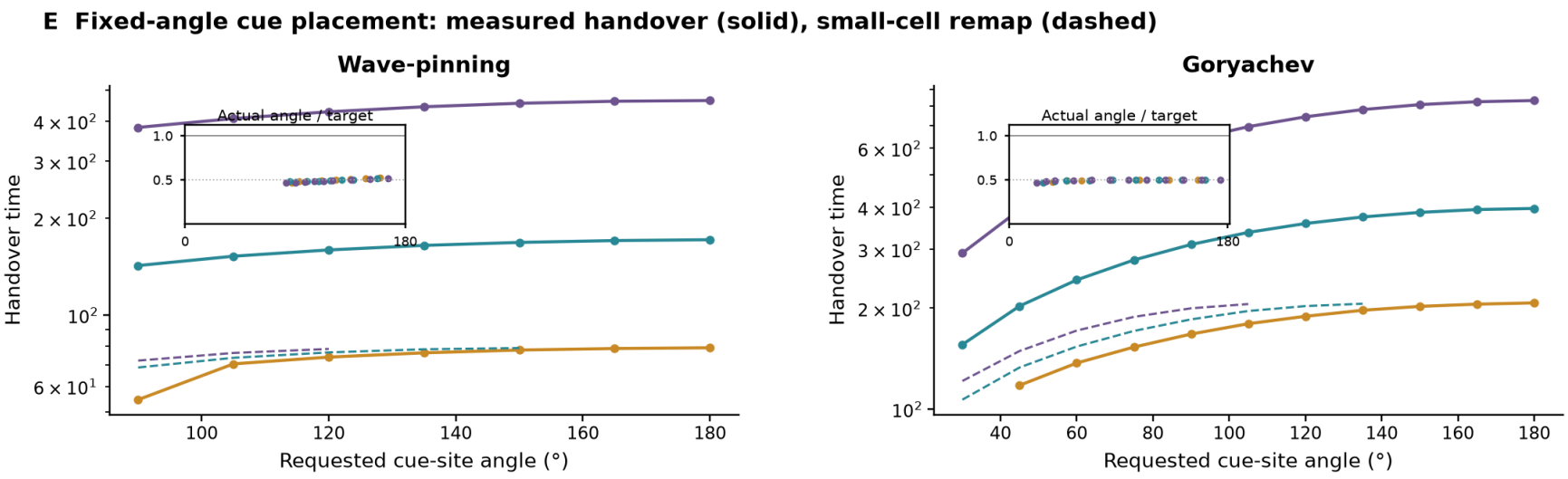


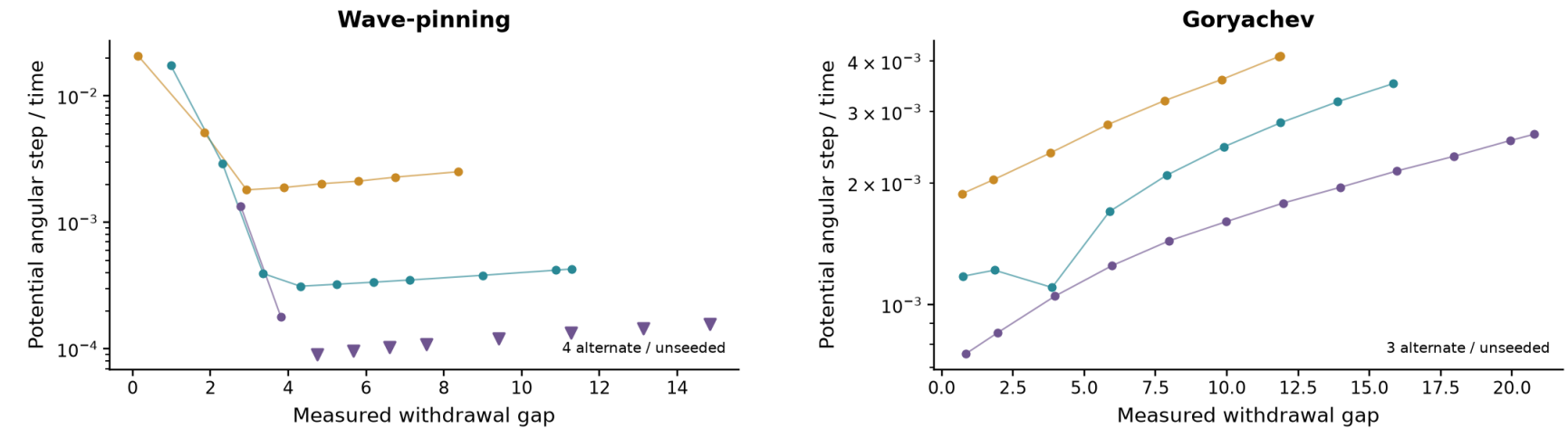
Fixed-size cue placement, native histories and separation responses

Figure 3. Geometry and measured activity direction · image 2 of 2

Figure 3 · Geometry remaps challenges; activity direction is measured separately







Dashed remaps stop at observed small-cell support. Insets measure actual global angle at handover, R≥0.05.
Potential step uses measured withdrawal centers and widths; direct heading-time tests and spatial outcomes are retained in the supplement.
These are native polarity-activity readouts; observed migration and conditional OU persistence estimates remain separate.

Measured-width geometry remaps and direct native-heading qualifications

Figure 3. Separation affects native competition differently across models

Test whether greater incumbent–challenger separation slows competition at fixed circumference, then examine its geometric translation into measured activity direction.

Detailed legend

A, A native cue is rotated at fixed size. The primary initial gap g runs from the incumbent initial 10%-contrast edge to the cue support near edge along the shorter route. Chemical circumferences are 30, 37.8 and 47.6; Spring resting-size labels are 2.5, 3.15 and 3.97, with actual circumference recorded. All 12 fresh single-region basals qualify without retuning. External placements alone enter the distance claim; borderline negative gaps use open symbols and internal references are supplementary.

B, Native Goryachev and Holmes 3 active-field histories at circumference 37.8 and external gaps 2 and 8. The dashed white line marks independently replayed complete-region handover. Holmes 3 uses active Rac. Chemical A1 histories complete 5, 000 or 20, 000 native units according to the retained continuation rule; Spring completes 2, 000. Every bound uses its actual completed horizon.

C1, Continuous-cue handover is the first time challenger excess active amount exceeds incumbent excess while two resolved, spatially associated regions exist. C2, Cue-free resolution follows seeding to complete-region challenger/incumbent ratio 1; all 0.5/1/2 results are supplementary. Its axis uses the measured current global 10%-contrast gap at withdrawal. Triangles retain completed finite-cap time lower bounds. Unseeded, overlapping-contour, merged and single-region outcomes remain explicit. Lines do not bridge unavailable, censored or topology-changing placements.

D, Requested center angle combines the physical edge gap, half the measured incumbent width and half the cue support, divided by circumference and expressed in radians. The relation is typeset in the main panel. A fixed edge gap is not automatically a fixed center displacement when native incumbent width changes with size. Current B widths, actual circumference, short/long routes and measured center displacement are retained.

E, Solid curves show measured handover time for requested cue angle; dashed curves evaluate the reference-size handover curve at larger-cell measured-width gaps. The geometry-only remap stops within observed reference support and does not interpolate through missing/censored neighbors. The inset directly measures actual global-heading angle at handover, typically about half the requested angle. Handover time is not time to complete the requested angular turn.

F, The fixed-physical-scale view divides measured region-center angular step by measured cue-free resolution time, with censoring propagated. This conditional potential rate is distinct from the directly observed vector-heading change. Full-angle target-time brackets,50/90/99% sensitivities, final headings, coherence coverage and cancellation gaps are supplementary.

Figure 3. Separation affects native competition differently across models · continued

Complete active amounts integrate native active concentration above a common instantaneous minimum over the two full valley-bounded basins, including signal outside the cue. Continuous boundary position uses a local three-point parabolic minimum; actual field amounts use exact piecewise-linear fractional-cell integration. Active flat/nonunique saddles are unscorable. Single moving regions and merged fields do not supply invented two-region handovers or competition times.

The global vector is the first spatial harmonic divided by instantaneous mean active field; $R = |M_1|/M_0$ is coherence, unlike Figure 4 basal-M1-normalized magnitude. Raw exact-direction availability uses $R > 10^{-10}$, with displayed $R \geq 0.05$ and 0.1/0.2 sensitivities. Direction-quality/cancellation gaps break angle unwrapping. All 1, 150 histories include saved-interval principal angular-rate time courses; these are not instantaneous RHS rates or counts of unobserved complete turns.

A2 uses 0.5, 0.75, 1, 1.5 and 2 times each placement own corrected handover, then 13, 000 cue-free chemical units or 1, 300 Spring units. B withdraws at first valid ratios 0.5, 1 or 2, then 20, 000 chemical units or 2, 000 Spring units. Native finite-field stationarity is reported independently of scalar/spatial fate. No finite cap substitutes for an unavailable own handover.

All native positivity and conserved-pool audits pass. All 24 B mirrors agree. A1 handover and B seed-time numerical differences are at most 0.0449%;35/36 B resolution brackets overlap. Goryachev C37.8, gap 4, ratio 1 shifts from 1040–1060 to 1080–1100 units on the finer mesh, about 4% in resolution timing. Native clocks differ; no absolute cross-model time ranking or cell-velocity law is imposed.

Narrative points and conclusions

- Under continuous forcing, Goryachev handover increases with separation, while Holmes 3 is nearly flat over valid two-region placements. Wave-pinning and Spring show near-edge increases followed by broad plateaux. These are model-dependent responses under the declared cue, not a universal absence of distance effects in Wave-pinning.
- Cue-free competition has a different distance response: Wave-pinning, Holmes 3 and Spring can coarsen very slowly, with many large-size resolution lower bounds. Goryachev includes a nonmonotone near-edge response at ratio 1; the full curve and every ratio are retained.
- Actual global-heading measurements qualify the proposed geometry-to-persistence inference. Complete-region handover typically occurs at about half the requested cue angle. Cue-free fields can resolve to an intermediate heading, so a potential center step is not always the realized angular event. These data do not establish an unconditional geometry-driven persistence gain.

Figure 3. Separation affects native competition differently across models · continued

Proposed measurements and experimental follow-up

At fixed native size, vary only cue placement and measure complete-region handover, incumbent 50% excess loss and challenger reaching 50% of basal incumbent excess.

Seed two complete regions to a declared excess ratio, withdraw the cue and measure their actual 10%-contrast separation, first single-peak resolution time, winner association and exact cue-free RHS exchange budget.

Measure the same-run global activity vector, its coherence, actual angular displacement, full-target time brackets and saved-interval angular rates; compare them with geometric remaps.

Scope, limitations and pending validation

Basal width, center displacement and edge gap differ; fixed edge gap is not automatically fixed physical center step across sizes.

A native activity-vector heading is not observed migration direction, a cell velocity law or a conditional OU persistence estimate.

Finite caps produce lower bounds, not proof of permanent arrest. Scalar/spatial winner and full native-field stationarity remain separate.

Two resolved peaks with merged primary 10% contours are ineligible for the primary B gap claim. One moving/merged region is a separate outcome.

The finer-mesh Goryachev B ratio 1 example has about 4% resolution-time sensitivity despite seed-time and winner agreement.

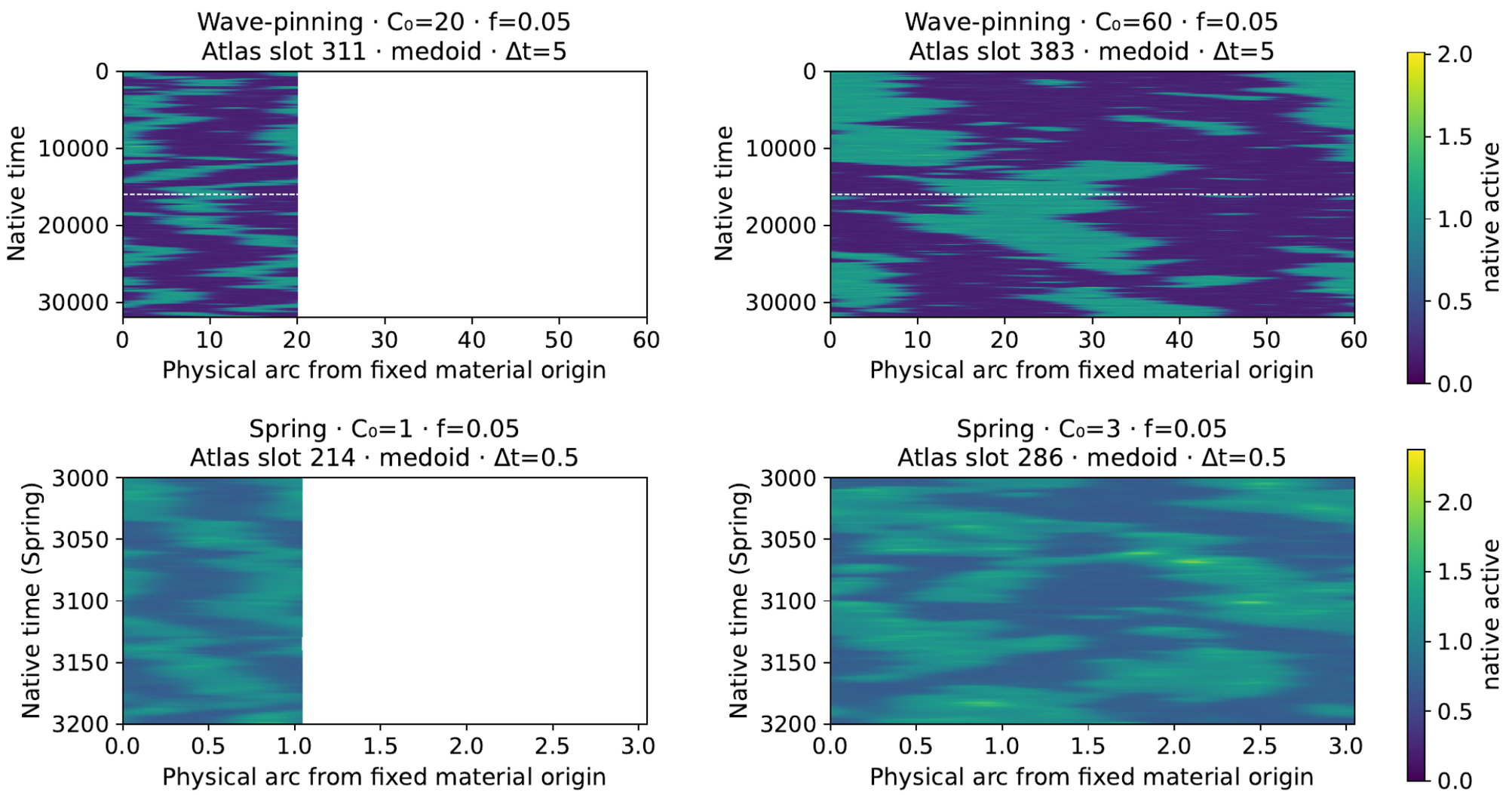
The native-pattern responses do not uniquely identify a molecular mechanism or provide causal allocation to one kinetic term. No noise, moving cue, dose sweep or deBelly ring enters this figure.

Sources and reproducibility

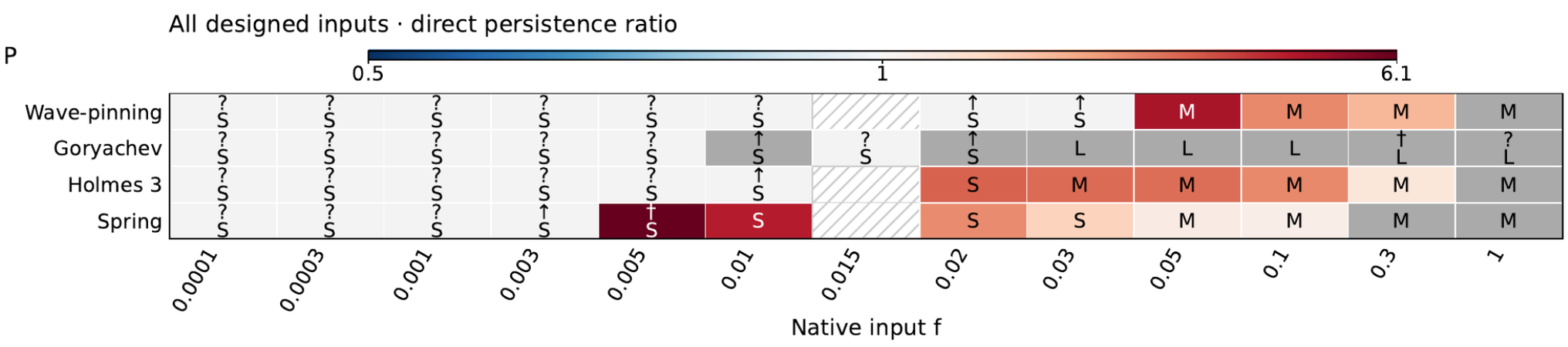
Complete corrected A1 evidence
All A2 switching brackets and unavailable calibrations
All B gap, fate, censoring and actual-heading measurements
Support-limited measured geometry remaps
Direct global-heading target checks
B numerical convergence including the material exception
Final artifacts and source-link audit
Lossless full native-history pixel audit

A size advantage in heading persistence can coexist with multiple active regions.

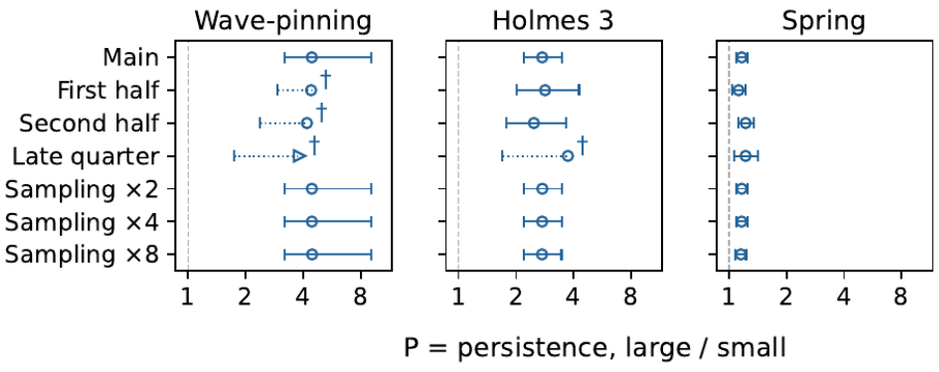
A



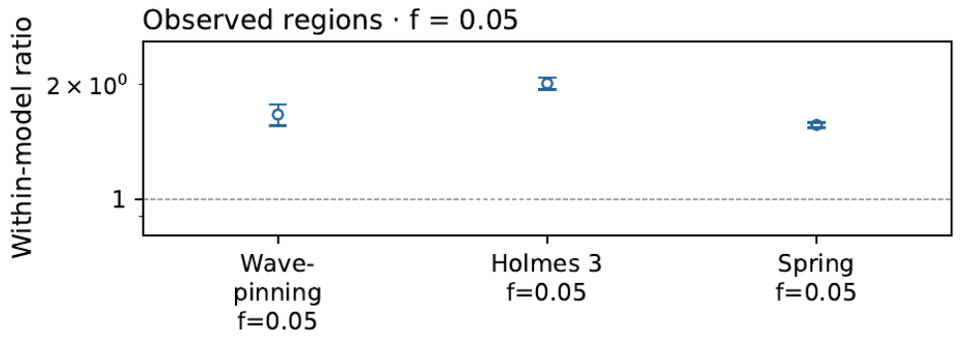
B



C Heading persistence · f = 0.05



D



A size advantage in heading persistence can coexist with multiple active regions.

A size advantage in heading persistence can coexist with multiple active regions.

Observed coexistence of a size advantage in activity-heading persistence with multiple active regions, with full hypothesis and qualification evidence.

Detailed legend

A. Native activity kymographs show the fixed, reviewed f=0.05 medoids for Wave-pinning C_0=20/60 and Spring C_0=1/3. Physical arc, native time and activity colour limits are shared within each model pair. Wave-pinning shows the full 0–32000 native horizon, with a white dashed line at the retained second-half analysis start. Spring shows the declared fixed final one-sixteenth viewport 3000–3200 for both sizes; its full 0–3200 histories remain in S11 and all statistics use the full declared analysis window. Full native grids and exact selected IDs are in the source inventories; concentrations are neither resampled nor quantized. Spring boundary geometry follows its actual circumference.

B. $P=\tau_p(\text{largest})/\tau_p(\text{smallest})$ uses direct heading persistence, with a logarithmic colour scale centred on one. The full grid retains all four models and every designed input. ? is unidentified or unavailable; arrows are one-sided empirical bounds; ↕ is a finite empirical interval; † marks an upper-unbounded bootstrap CI. Bound cells have no fabricated ratio colour. S/M/L/X denote the largest-cell native single-front/multiple-front/low-contrast/mixed regime. Grey retains operational forcing; hatching marks an input not designed for that model.

C. Direct persistence ratios at f=0.05 are shown for all seven declared windows/cadences. Circles are finite ratios; open right-pointing triangles are empirical lower bounds. Horizontal intervals and vertical caps show whole-seed 95% bootstrap uncertainty. A dagger is an unbounded CI upper endpoint; its finite lower endpoint remains distinct from the empirical bound. All three models have a positive lower confidence bound above one in every shown view. Sampling variants reuse the same independent histories.

D. Primary optical region-count quotients compare large/small within each model at f=0.05. The imaging proxy uses fixed physical Gaussian smoothing 1.5 (Spring 0.15), threshold 0.5 and native cadence 80 (Spring 8). Caps retain the exact whole-seed 95% intervals, including Spring’s narrow interval. These counts describe observed local organization and are not a validated predictor of persistence.

The selected f=0.05 conditions were not selected for E4 extension. Their common and longest-arm tables reuse the same common-horizon cohort; agreement between those arms is not independent long-horizon corroboration. Each selected size has 12 independent noisy seeds. Native clocks differ between chemical models and Spring; no cross-model absolute time or amplitude ranking is implied.

Figure 4 R is the native first-mode magnitude divided by its exact no-input basal first-mode magnitude. It can exceed one and differs from Figure 3 coherence |M1|/M0. Global-heading-derived physical-mobility proxies are distinct from qualified single-region literal site MSDs.

A size advantage in heading persistence can coexist with multiple active regions. · continued

Narrative points and conclusions

- At f=0.05, direct large/small heading-persistence ratios are 4.44 for Wave-pinning,2.74 for Holmes 3 and 1.17 for Spring. All seven declared views retain a positive lower confidence bound above one. The largest-size native multiple-front fractions are 0.639,0.857 and 0.659 respectively.
- The complete predeclared hypotheses remain qualified: H1 is inconclusive, H2 does not meet its overall criterion and H3 is inconclusive in both primary observation arms. The full S0 memo shows every model, deciding numerical shortfall and two alternative framings. A6 does not satisfy its declared prediction-support gate; increased multiplicity is descriptive.
- These one-dimensional polarity observables motivate experimentally measurable local activity and heading assays. They do not establish a cell-velocity law or equate simulated heading persistence with observed migration persistence.

Proposed measurements and experimental follow-up

Common observations use native chemical 32000 and Spring 3200 horizons. Longest observations use the frozen per-condition selector, including chemical 128000 or Spring 12800 where E4 was preselected. Mixed per-size horizons remain explicit; raw total-count quotients are unavailable when exposures differ.

Persistence and its empirical bounds come from measured heading correlations with declared coverage. Bootstrap uncertainty resamples whole independent seed histories. Deterministic matched-mean/zero controls have 0 independent stochastic seeds and 0 seed-bootstrap resamples.

Primary optical counts and strict dominant-newborn takeovers use complete observed-region amounts. A takeover requires the newborn to be largest among qualified regions and its pairwise D against the previous dominant region to exceed 0.5 for two frames. With three or more regions this does not mean 75% of whole-cell activity.

Spring forcing is material attached, with physical correlation length 0.15 C/C_basal; fixed physical optical smoothing is separate. Literal A3 positions use the fixed material-origin angle times actual circumference, splitting unresolved antipodal edges; paired ±π routes are deterministic sensitivities.

Scope, limitations and pending validation

H1 is inconclusive, H2 fails the overall criterion and H3 is inconclusive; the positive selected comparison is not support for those full hypotheses.

A size advantage in heading persistence can coexist with multiple active regions. · continued

The three-model selected comparison uses 12 independent seeds per size; Goryachev remains visible in the full overview and is forced/undercovered at this input.

Late-quarter Wave-pinning/Holmes persistence ratios are bounded, not finite point estimates; unbounded bootstrap upper endpoints must remain visible.

The persistence direction and native majority-regime labels pass selected A7 checks; short-lag mobility itself is cadence-sensitive in some selected conditions and is not promoted as a robust headline.

Primary optical counts are imaging proxies at threshold 0.5, fixed physical smoothing 1.5/0.15 and cadence 80/8 in native units; native regime fractions remain distinct.

Heading persistence and local activity regions are simulated polarity observables, not measured migration persistence or a cell-velocity law.

Geometric qualifications and literal single-region site MSD remain in S4/S5; the global-heading-derived physical-mobility proxy is not one tracked site diffusivity.

Historical Spring sampled-duration/cadence failures, finite-window bounds, coverage exclusions and E4 source-recovery/original-cohort distinctions remain explicit.

Short-lag global-heading physical-mobility estimates may increase while angular persistence increases. In multiple-region states they do not measure diffusion of one tracked site. Selected mobility estimates have cadence sensitivity and are excluded from the headline.

E4 native source repair is an engineering acceptance under unchanged tolerances, not a stationarity/persistence conclusion. Retained prefixes, sigma branches, route choices, numerical companions and common/longest references are not extra independent seed replicates.

All 578 required native-atlas pages are retained in 13 numbered lossless appendix volumes. The core PDF is the statistical supplement and appendix index. Full 17-significant-digit native CSV.gz fields remain verified in the HPC archive; every page has portable exact-source inventories and compact companions.

For Spring C0=3, f=1, seed202610192320800, the original full numerical gate failed 12 scalar-observation cadence comparisons. The accepted stable scope passes 180 rows and excludes all 18 q90 multiregion-span rows, including six passing ODE-only comparisons. These are consecutive saved multiregion-mask durations, not tracked individual lifetimes. The full-window q90 changes from 10 to 7.15 when scalar cadence changes from 0.5 to 0.05. Exact sigma-one replay preserves an existing path and forcing; it does not establish numerical or observation convergence. Numerical companions and dense samples are not independent scientific seeds.

A size advantage in heading persistence can coexist with multiple active regions. · continued

Sources and reproducibility

E0
E1
E2
E3
E4
A1
A2
A3
A4
A5
A6
A7
windows_and_coverage
common_horizon
long_horizon
framing
independent_final_decision_audit
final_statistics_producer
S11_full_atlas_coverage
reviewed_interpretation_memo
reviewed_interpretation_memo_display
main_display_audit
Historical Spring observation-cadence qualification
Independent Figure 4 completion review

Supplementary material and supporting analyses

Figure 2: complete revised evidence

The comprehensive supplement retains the full designed condition grid, unavailable measurements, finite-horizon bounds, native histories, controls and the stated numerical and observational sensitivities. Earlier figure versions remain development history.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/signature_assays_20261020/supporting/vectors/figure2_signature_assays_complete_supplement.pdf

Figure reader and source data

Independent completion review

Figure 3: complete revised evidence

The comprehensive supplement retains the full designed condition grid, unavailable measurements, finite-horizon bounds, native histories, controls and the stated numerical and observational sensitivities. Earlier figure versions remain development history.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261020/figure3_distance/supporting/vectors/figure3_distance_supplement.pdf

Figure reader and source data

Independent completion review

Figure 4: complete revised evidence

The comprehensive supplement retains the full designed condition grid, unavailable measurements, finite-horizon bounds, native histories, controls and the stated numerical and observational sensitivities. Earlier figure versions remain development history.
S0–S12 statistical core and complete native-atlas appendix index; native pages are in all linked appendix volumes. Every numbered appendix below is part of the complete supplement.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/signature_figure4_20261020/supporting/vectors/figure4_revision_complete_supplement.pdf

Figure reader and source data

Independent completion review

S11 native atlas appendix 001

S11 native atlas appendix 002

S11 native atlas appendix 003

S11 native atlas appendix 004

S11 native atlas appendix 005

S11 native atlas appendix 006

S11 native atlas appendix 007

S11 native atlas appendix 008

S11 native atlas appendix 009

S11 native atlas appendix 010

S11 native atlas appendix 011

Supplementary material and supporting analyses · continued

S11 native atlas appendix 012

S11 native atlas appendix 013

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope
Figures 2–4 revision specification
Figure 2 October 20 revision brief
Figure 3 October 20 revision brief
Figure 4 October 20 revision brief